	Course Code:B20BS2104			
	SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R20**
	II B. Tech I Semester - MODEL QUESTION PAPER			
	APPLIED MATHEMATICAL TECHNIQUES			
	Computer Science & Business System (CSBS)			
TIME: 3 Hrs.				**Max. Marks: 70 M**
	Answer any One Question from Each Unit			
	All questions carry equal marks			
	UNIT-I	**M**	**CO**	**KL**
1.a)	Determine a Fourier series to represent $x - x^2 from x = -\pi to x = \pi$	7	CO1	K3
b)	Determine the Fourier series for the function $f(x) = \begin{cases} \pi x, & 0 \le x < 1 \\ \pi(2-x), & 1 \le x \le 2 \end{cases}$ and deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots\ldots\ldots = \frac{\pi^2}{8}$	7	CO1	K3
	OR			
2. a)	Determine the half range cosine series for the function $f(x) = (x-1)^2 \, in \, 0 < x < 1$ and hence show that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\ldots\ldots\ldots = \frac{\pi^2}{6}$	7	CO1	K3
b)	Determine the sine series for $f(x) = x in \, 0 \le x \le \pi$	7	CO1	K3
	UNIT-II			
3.a)	Using Fourier Integral establish $\int_0^\infty \frac{\omega Sin\omega x}{1+\omega^2} d\omega = \frac{\pi}{12} e^{-x} (x > 0)$	7	CO2	K3
b)	Determine the Fourier transform of $f(x) = \begin{cases} 1 & for\|x\| < 1 \\ 0 & for\|x\| > 1 \end{cases}$. Hence evaluate $\int_0^\infty \frac{Sinx}{x} dx$	7	CO2	K3
	OR			
4. a)	Determine the Fourier sine transform of $\frac{e^{-ax}}{x}$	7	CO2	K3
b)	Using Parsevals identity show that $\int_0^\infty \frac{dt}{(t^2+1)(t^2+4)} = \frac{\pi}{12}$	7	CO2	K3
	UNIT-III			
5.a)	Develop partial differential equations by eliminating the arbitrary functions from (i) z = y f(x) + x g(y) and (ii) xyz = F(x + y + z).	7	CO3	K3
b)	Solve the equation x(y-z) p + y(z-x) q = z(x-y) with usual notation**.**	7	CO3	K3
	OR			
6. a)	Solve the partial differential equation $(D^3 + D^2D' - DD'^2 - D'^3) z = e^x.\cos(2y)$ with usual notation.	7	CO4	K3
b)	Solve the partial differential equation $(D - D' - 1)(D - D' - 2) z = e^{2x - y}$ with usual notation.	7	CO4	K3
	UNIT-IV			
7. a)	Using Newton's forward interpolation formula, for the given table of values (see table below)	7	CO5	K3

X	1.1	1.3	1.5	1.7	1.9
$f(x)$	0.21	0.69	1.25	1.89	2.61

	Obtain the value of f(x) when x= 1.4.			
b)	Using Lagrange formula, calculate f(3) from the following table <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>4</td><td>5</td><td>6</td></tr><tr><td>f(x)</td><td>1</td><td>14</td><td>15</td><td>5</td><td>6</td><td>19</td></tr></table>	7	CO5	K3
	OR			
8. a)	Determine a real root of an equation $\cos x + 1 - 3x = 0$ by using False position method	7	CO5	K3
b)	Derive a formula to find the cube root of N using Newton-Raphson method and hence find the cube root of 18.	7	CO5	K3
	UNIT-V			
9.a)	The following table gives the velocities of a car, moving on a straight road at intervals of two minutes. Determine the displacement of the car by the use of Trapezoidal rule. <table><tr><td>Time (in mts)</td><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td></tr><tr><td>Vels (km/hr)</td><td>0</td><td>20</td><td>30</td><td>40</td><td>25</td><td>15</td><td>0</td></tr></table>	7	CO6	K3
b)	Estimate the numerical value of $\int_0^1 \frac{1}{1+x} dx$ by the use of Simpson's 1/3 rule taking six subintervals. Compare with the value obtained by direct integration	7	CO6	K3
	OR			
10.a)	Solve $\frac{dy}{dx} = x\sqrt{1+y^2}, y(1) = 0$ at x=3 taking h=0.5 using Euler's method.	7	CO6	K3
b)	Determine y(0.1) , y(0.2), and y(0.3) using Runge-Kutta fourth order formula given that $\frac{dy}{dx} = x + x^2y^2, y(0) = 1$.	7	CO6	K3

Course Code:B20CB2101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				**R20**	
II B. Tech I Semester - MODEL QUESTION PAPER					
FORMAL LANGUAGES AND AUTOMATA THEORY					
Computer Science & Business System (CSBS)					
TIME: 3 Hrs.				**Max. Marks: 70 M**	
Answer any One Question from Each Unit					
All questions carry equal marks					
UNIT-I			**M**	**CO**	**KL**
1.	**(a)**	Define i) Alphabet ii) String iii) Language iv) Kleen Closure.	**4**	**CO1**	**K2**
	(b)	Design a DFA which accepts even number of 0's and 1's over alphabet ∑ ={0,1} , with neat transition table and diagram	**10**	**CO1**	**K3**
(OR)					
2.	**(a)**	Let L1 = {010}, L2={01,0} then find (i) L1L2 (ii) L1 * (iii) L2 + (iv) L1 * + L2 *	**7**	**CO1**	**K3**
	(b)	Define Moore and Mealy machines with examples	**7**	**CO1**	**K2**
UNIT-II					
3.	Define DFA and RE. Design a DFA that accepts all strings formed from the RE 1*01(0+11)*. Also explain how to convert a regular expression to DFA.		**14**	**CO1**	**K4**
(OR)					
4.	Convert the following regular expressions to NFA with epsilon transitions a) 0*+110 b) (0+1)*		**14**	**CO1**	**K4**
UNIT-III					
5.	Explain the following with examples i) Left Most Derivation ii) Right Most Derivation iii) Ambiguous grammar iv) Unit Production v) Null Production.		**14**	**CO2**	**K2**
(OR)					
6.	**(a)**	Define CFG and explain different types of Normal Forms in CFG with Examples.	**7**	**CO2**	**K2**
	(b)	Convert the following productions into GNF $\mathbf{A_1 \rightarrow A_2A3}$, $\mathbf{A_2 \rightarrow A_3A_1 /b}$, $\mathbf{A_3 \rightarrow A_1A_2 /a}$	**7**	**CO2**	**K3**
UNIT-IV					
7.	**(a)**	Define PDA and write about its acceptance of a Language.	**4**	**CO2**	**K2**
	(b)	Construct PDA for the language L={ $a^n b^n$ / n>=1}	**10**	**CO2**	**K3**
(OR)					
8.	**(a)**	Define Left Most and Right Most Derivations in CFG.	**6**	**CO2**	**K2**
	(b)	Design a **PDA** which accepts Even length palindrome strings formed over input alphabet Σ∈ {a,b}.	**8**	**CO2**	**K3**
UNIT-V					
9.	**(a)**	Define Universal Turing Machine.	**4**	**CO3**	**K2**
	(b)	Design a Turing Machine for Language **L={$a^n b^n c^n$ /n>=1}**	**10**	**CO3**	**K3**
(OR)					
10.	**(a)**	Write about Church Thesis.	**7M**	**CO3**	**K2**
	(b)	Define Post Correspondence Problem and Find whether the lists **M = (abb, aa, aaa) and N = (bba, aaa, aa)** have a Post Correspondence Solution?	**8M**	**CO3**	**K2**

Course Code:B20CB2102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				**R 20**	
II B. Tech. I Semester Examinations(MODEL QUESTION PAPER)					
COMPUTER ORGANIZATION & ARCHITECTURE					
Computer Science & Business System (CSBS)					
Time: 3 Hrs.					**Max. Marks: 70**
Answer any one question from each unit					
All questions carry equal marks					
			CO	**KL**	**M**
UNIT-I					
1.	(a).	Describe Fixed Point Representation	CO1	K2	7M
	(b).	Explain Addition and Subtraction algorithm	CO1	K2	7M
(OR)					
2.	(a).	Describe Bus Structures	CO1	K2	7M
	(b).	Explain Division Algorithm with example	CO1	K2	7M
UNIT-II					
3.	(a).	Describe Instruction cycle in computer system	CO1	K2	7M
	(b).	Explain RTL and Register Transfer	CO1	K2	7M
(OR)					
4.	(a).	Hardwired control Vs Micro programmed control	CO1	K2	7M
	(b).	Explain about Computer instructions	CO1	K2	7M
UNIT-III					
5.	(a).	Explain about General register organization with seven registers	CO2	K2	7M
	(b).	Expand the given statement in Three, Two, One ,Zero Addresses A=(B+C)*(D/E*F)	CO2	K3	7M
(OR)					
6.	(a).	Describe the Addressing Modes i) Direct ii) Relative iii) Auto Increment	CO2	K2	7M
	(b).	List out the Characteristics of RISC and CISC	CO2	K1	7M
UNIT-IV					
7.	(a).	Discuss about Virtual memory	CO3	K2	7M
	(b).	Explain about Asynchronous Communication interface with neat diagram	CO3	K2	7M
(OR)					
8.	(a).	Explain about Handshaking Data transfer	CO3	K2	7M
	(b).	Explain Associative Memory	CO3	K2	7M
UNIT-V					
9.	(a).	Describe the Characteristics of Multiprocessors	CO4	K2	7M
	(b).	Explain about Parallel processor	CO4	K2	7M
(OR)					
10.	(a).	Describe Interconnection Structures with neat diagram	CO4	K2	7M
	(b).	Describe about Instruction Pipeline	CO4	K2	7M

Course Code:B20CB2103

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) **R 20**

II B. Tech. I Semester Examinations(MODEL QUESTION PAPER)

DATABASE MANAGEMENT SYSTEMS

Computer Science & Business System (CSBS)

Time: 3 Hrs. **Max. Marks: 70**

Answer any one question from each unit

All questions carry equal marks

			M	**CO**	**KL**
		UNIT-I			
1	a.	Distinguish between file system versus Database Management System	7	1	K2
	b.	Explain different levels of abstraction offered by DBMS with an example	7	1	K2
		OR			
2	a.	Draw a neat diagram of the structure of DBMS and explain the functions of various components of DBMS.	8	1	K2
	b.	Explain 2-tier and 3-tier architecture of DBMS	6	1	K2
		UNIT-II			
3	a.	Write an E-R diagram for University database	8	2	K3
	b.	Explain with an example specialization, generalization and aggregation	6	2	K2
		OR			
4	a.	Consider the following tables: Students(Std_ID:String, S_Name:String:, Date_of_Birth: Date) and Enrolled(Std_ID String, Course_ID String, Grade Char(1)) Answer the following queries in SQL. i) Write DDL statement to create Students table. ii) Insert the values into the table.	7	2	K3
	b.	Consider an example of your own how an E-R diagram with one-many converted into relational tables.	7	2	K3
		UNIT-III			
5	a.	Consider the following tables: Suppliers(Supp_ID: Integer, S_Name: String, Address:String) Parts(P_ID: Integer, Pname: String, Colour: String) Catalog(Supp_ID: Integer, P_ID: Integer, Cost: Real). Answer the following queries in SQL. i) Write DDL statements to create Suppliers and Catalog tables with key constraints and referential constraint. ii) Find IDs and names of Suppliers who supply some "Red" colour part. iii) Find the details of suppliers whose names are 5 characters long and begin with character 'A'.	6	3	K3
	b.	Explain various join operations in SQL	8	3	K3
		OR			
6	a.	Define Query Processing? Explain with neat diagram various steps involved in query processing and optimization	7	3	K3
	b.	Explain with an example how to evaluate relational algebraic expression	7	3	K3
		UNIT-IV			

7	a.	Consider the schema R(A,B,C,D,E,G) and the list of functional dependencies F={A→BC, EC→D,D→A, G→E }. Determine all candidate keys of R.	7	4	K4
	b.	Explain with an example join dependency and fifth normal form	7	4	K3
OR					
8	a.	Find the highest normal form satisfied by the relation R(A,B,C,D,E) with set of FDs F= {BC->D, AC->BE, B->E}	7	4	K4
	b.	Prove that a relation in BCNF is also in 3 NF	7	4	K3
UNIT-V					
9	a.	Define a Transaction? Explain ACID Properties	7	5	K2
	b.	Explain Two-Phase locking Protocol?	7	5	K2
OR					
10	a.	Define Access Control? Explain various types of Access Controls?	7	5	K2
	b.	Explain Hash based Indexing	7	5	K2

					Course Code:B20CB2104
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R20**
II B. Tech I Semester - MODEL QUESTION PAPER					
OBJECT ORIENTED PROGRAMMING THROUGH JAVA					
Computer Science & Business System (CSBS)					
TIME: 3 Hrs.					**Max. Marks: 70 M**
Answer any One Question from Each Unit					
All questions carry equal marks					
			CO	**KL**	**M**
UNIT - I					
1.	**a).**	Illustrate the concept of JVM with a Diagram.	**CO1**	**K3**	**7M**
	b).	Illustrate the differences between C,C++ and Java with a neat diagram	**CO1**	K3	7M
OR					
2.	**a).**	Illustrate the structure of a java program	**CO1**	**K3**	**7M**
	b).	Explain about java buzz words	**CO1**	**K3**	**7M**
UNIT – II					
3.	**a).**	Write a java program to perform constructor overloading with an example	**CO2**	**K2**	**7M**
	b).	Explain class Declaration syntax in java and modifiers	**CO2**	**K2**	**7M**
OR					
4.	**a).**	Write a java program to perform method overloading.	**CO2**	**K2**	**7M**
	b).	Explain the keyword final how it is used in java	**CO2**	**K2**	**7M**
UNIT – III					
5.	**a).**	Explain polymorphism and its types. Construct a java program which Illustrates the functionality of method overloading and method overriding.	**CO3**	**K3**	**7M**
	b).	Write how Multiple Inheritance is possible in java with an example program.	**CO3**	**K3**	**7M**
OR					
6.	**a).**	Demonstrate an array? Write a java program to read an array of **n** elements and print them.	**CO3**	**K3**	**7M**
	b).	Write a java program to find second largest number in an array.	**CO3**	**K3**	**7M**
UNIT – IV					
7.	**a).**	Interpret the concept of packages in java.	**CO4**	**K2**	**7M**
	b).	Construct a java program that shows the functionality of creating a public class in an already existing user defined package.	**CO4**	**K2**	**7M**
OR					
8.	**a).**	Write a java program to create custom exception.	**CO4**	**K3**	**7M**
	b).	Illustrate 2 different ways of creating a thread in java with code.	**CO4**	**K3**	**7M**
UNIT – V					
9.	**a).**	Explain any 4 layout mangers in java with example code.	**CO5**	**K2**	**7M**
	b).	Write a java swing code to create a frame and add any five different components onto the frame.	**CO5**	**K3**	**7M**

OR					
10.	**a).**	Write a java program to explain event handling.	**CO5**	**K2**	**7M**
	b).	Explain difference between Swings and AWT	**CO5**	**K2**	**7M**

					Course Code:B20CB2201
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R20**
II B. Tech II Semester - MODEL QUESTION PAPER					
OPERATING SYSTEMS					
Computer Science & Business System (CSBS)					
TIME: 3 Hrs.					**Max. Marks: 70 M**
Answer any One Question from Each Unit					
All questions carry equal marks					
			CO	**KL**	**M**
UNIT – I					
1.	a).	List and Describe the major services provided by Operating System	1	K2	7
	b).	Why Micro Kernel Structure is better than Layered Structure? Explain.	1	K2	7
OR					
2.	a).	Discuss the purpose of Interrupts? How the Interrupts are handled by Operating System?	1	K2	7
	b).	What is Debugging? Explain OS debugging?	1	K2	7
UNIT – II					
3.	a).	The following table represents details of four processes Process / Arrival Time / Burst Time P1 / 0 / 6 P2 / 4 / 8 P3 / 6 / 7 P4 / 2 / 3 Calculate average waiting time and Average Turnaround time using the following scheduling algorithms i) Non-Preemptive Shortest Job First Scheduling algorithm. ii) Preemptive Shortest Job First Scheduling algorithm	2	K3	8
	b).	Explain any two Multi Processor scheduling algorithms? List out its advantages and limitations	2	K2	6
OR					
4.	a).	The following table represents details of four processes Process / Arrival Times / Burst Time / Priority P1 / 8 / 9 / 4 P2 / 6 / 5 / 5 P3 / 4 / 6 / 1 P4 / 4 / 5 / 6 P5 / 0 / 6 / 3 P6 / 2 / 8 / 2 Calculate average waiting time and Average Turnaround time using the following scheduling algorithms i) Non-Preemptive Priority scheduling algorithm ii) Preemptive Priority scheduling algorithm	2	K3	8
	b).	Identify the different issues in thread scheduled?	2	K2	6
UNIT – III					
5.	a).	Explain the following terms with suitable examples i) Critical Section	3	K2	6

		ii) Semaphore			
	b).	Design a solution for handling deadlock when it is occurred?	3	K3	8
OR					
6.	a).	Write a semaphore solution for the Readers and Writers problem	3	K3	7
	b).	Design a deadlock avoidance algorithm? Outline the advantages and Limitations.	3	K3	7
UNIT – IV					
7.	a).	Summarize different Free space management techniques used in contiguous memory allocation?	4	K2	7
	b).	What is Inverted Paging? How it different from paging?	4	K2	8
OR					
8.	a).	Find out number of page faults for the given page requests using Least recently Used page replacement algorithm when i) Number of frames =3 ii) Number of Frames = 4 Where, page requests: 1,3,4,0,5,3,2,1,0,4,5,2	4	K3	8
	b).	What is paged segmentation? How it is different from paging?	4	K2	6
UNIT – V					
9.	a).	Consider a disk queue contains cylinder requests for I/O are: 98, 183, 41, 122, 14, 124, 65, 67. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Find out head movement (in number of cylinders) incurred while servicing these requests using i) First Come First Serve disk arm scheduling algorithm. ii) Shortest Seek Time First disk arm scheduling algorithm.	4	K3	8
	b).	What is system protection? Explain different protection mechanisms?	4	K2	6
OR					
10.	a).	What is file system implementation? Explain i-node file allocation algorithm?	4	K2	7
	b).	How the Processes and Files are managed in Windows operating system?	4	K2	7

Course Code:B20CB2202					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R20**
II B. Tech II Semester - MODEL QUESTION PAPER					
PYTHON PROGRAMMING					
Computer Science & Business System (CSBS)					
TIME: 3 Hrs.			**Max. Marks: 70 M**		
Answer any One Question from Each Unit					
All questions carry equal marks					
			CO	**KL**	**M**
UNIT-I					
1	a)	Explain about input validation loops and nested loops with examples	CO1	K3	7
	b)	Write a Python program to calculate the amount payable if money has been lent on simple interest. Principal or money lent = P, Rate of interest = R% per annum and Time = T years. Then Simple Interest (SI) = (P x R x T)/ 100. Amount payable = Principal + SI. P, R and T are given as input to the program.	CO1	K4	7
OR					
2	a)	Explain about explicit conversion with examples.	CO1	K2	7
	b)	Explain about precedence of all operators in Python.	CO1	K2	7
UNIT-II					
3	a)	Define Python string padding functions? Explain with examples	CO2	K2	7
	b)	Illustrate if, if-else, if-elif-else Statements with examples.	CO2	K3	7
OR					
4	a)	Explain about data encryption in Python.	CO2	K2	7
	b)	Presume that a ladder is put upright against a wall. Let variables length and angle store the length of the ladder and the angle that it forms with the ground as it leans against the wall. Write a Python program to compute the height reached by the ladder on the wall for the following values of length and angle: i) 16 feet and 75 degrees ii) 20 feet and 0 degrees iii) 24 feet and 45 degrees iv) 24 feet and 80 degrees	CO2	K3	7
UNIT-III					
5	a)	Write a Python program to create three dictionaries, then create one dictionary that will contain the other three dictionaries.	CO3	K3	7
	b)	Describe Python list/Array methods? Explain.	CO3	K2	7
OR					
6	a)	Discuss about importing module from a package.	CO3	K2	7
	b)	Explain about anonymous or Lambda function with merits and demerits	CO3	K2	7
UNIT-IV					
7	a)	Explain about structuring classes with inheritance and polymorphism.	CO4	K2	7
	b)	Illustrate manipulating file pointer using seek with suitable example.	CO4	K3	7
OR					
8	a)	Demonstrate the case study of an ATM using classes.	CO4	K3	7

	b)	Explain about reading numbers from a file using Python program.	CO4	K2	7
UNIT-V					
9	a)	Defining clean-up actions.	CO5	K2	7
	b)	Illustrate Entry fields for the input and output of text with example.	CO5	K3	7
OR					
10	a)	Describe user Defined exception with example.	CO5	K2	7
	b)	Define Scrolling list boxes With example.	CO5	K2	7

					Course Code:B20CB2203
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
II B. Tech II Semester - MODEL QUESTION PAPER					
DESIGN AND ANALYSIS OF ALGORITHMS					
Computer Science & Business System (CSBS)					
TIME: 3 Hrs.					**Max. Marks: 70 M**
Answer any One Question from Each Unit					
All questions carry equal marks					
			CO	**KL**	**M**
UNIT-I					
1	a)	Define algorithm. Explain asymptotic notations Big O, Omega, and Theta.	CO1	K3	7
	b)	Write an algorithm for matrix multiplication and find its time complexity.	CO1	K2	7
OR					
2	a)	What is an articulation point? Explain the procedure to determine bi-connected components in the graph with example.	CO1	K5	7
	b)	Write an algorithm for BFS. Explain with example.	CO1	K4	7
UNIT-II					
3	a)	Explain divide-and-conquer technique. Write a recursive algorithm for finding the maximum and minimum element from the list.	CO2	K4	7
	b)	Illustrate the tracing of Quick Sort algorithm for the following set of numbers 25, 10, 72, 18, 40, 11, 64, 58, 32, 9	CO2	K4	7
OR					
4	a)	Find the optimal solution of the knapsack instance n = 7, M = 15, (p1, p2, …p7) = (10, 5, 15, 7, 6, 18, 3) and (w1, w2, ……w7) = (2, 3, 5, 7, 1, 4, 1).	CO2	K2	7
	b)	What is job sequencing with deadlines problem? Let n=5, profit (10, 3, 33, 11, 40) and deadlines (3, 1, 1, 2, 2) respectively. Find the optimal solution using greedy method.	CO2	K3	7
UNIT-III					
5	a)	For the given cost matrix, obtain an optimal cost tour using dynamic programming 0 10 15 20 5 0 9 10 6 13 0 12 8 8 9 0	CO3	K3	7
	b)	What is graph coloring? Write an algorithm for it and explain with an example	CO3	K2	7
OR					
6	a)	Using dynamic programming, solve the following knapsack instance n=4, m=5, (W1, W2, W3, W4)=(2, 1, 3, 2), (P1, P2, P3, P4)=(12, 10, 20, 15).	CO3	K2	7
	b)	Explain Multistage graphs with example. Write multistage graph algorithm to forward approach.	CO3	K4	7
UNIT-IV					

7	a)	What is backtracking? Apply backtracking to solve the instance of the sum of subset problem n=6, d=30, S={5, 10, 12, 13, 15, 18}	CO4	K4	7
	b)	Explain backtracking concept. Illustrate N queens problem using backtracking to solve 4-Queens problem.	CO4	K4	7
OR					
8	a)	State travelling salesperson problem. Apply Branch and Bound algorithm to solve the TSP instantiated by the following cost matrix. ∞20 30 10 11 15 ∞16 4 2 3 5 ∞2 4 19 6 18 ∞3 16 4 7 16∞	CO4	K4	7
	b)	Explain FIFO Branch and Bound	CO4	K2	7
UNIT-V					
9	a)	Explain the classes of NP-Hard and NP-complete.	CO5	K4	7
	b)	State and prove Cook's theorem.	CO5	K2	7
OR					
10	a)	Apply the Rabin-Karp algorithm to search for the pattern AABA in the text A A B A A C A A D A A B A A B A	CO5	K3	7
	b)	Apply the Knuth-Morris-Pratt matching to search for the pattern BAOBAB in the text BESS_KNEW_ABOUT_BAOBABS	CO5	K3	7

				Course Code:B20HS2202
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				**R20**
II B. Tech II Semester - MODEL QUESTION PAPER				
FUNDAMENTALS OF MANAGEMENT				
Computer Science & Business System (CSBS)				
TIME: 3 Hrs.				**Max. Marks: 70 M**
Answer any One Question from Each Unit				
All questions carry equal marks				
		CO	**KL**	**M**
	UNIT-I			
1.	A. Define Management and Explain thenature of Management	**CO1**	**K2**	07
	B. Write about the Significance of Management	**C01**	**K2**	**07**
	OR			
2.	A. Explain the principles of Management as outlined by Henry Fayol	**CO1**	**K2**	07
	B. Describe the functions of Management	**CO1**	**K2**	**07**
	UNIT-II			
3.	A. Define HRM & Differentiate HRM & Personnel Management	**CO2**	**K2**	07
	B. Describe about Training & Development	**CO2**	**K2**	**07**
	OR			
4.	A. Define Marketing, Explain in detail about Marketing mix	**CO2**	**K2**	07
	B. What are the stages of Product Life Cycle	**C02**	**K2**	**07**
	UNIT-III			

5.	A. Define Capital? Explain about types of Capital	**CO3**	**K2**	07
	B. Classify short-term & medium-term sources of Capital	**CO3**	**K2**	**07**
	OR			
6.	A. What are the elements of Corporate Planning Process	**CO3**	**K2**	07
	B. What do you Mean by SWOT analysis?	**CO3**	**K2**	**07**
	UNIT-IV			
7.	A. Write about Attitude and Perception	**CO4**	**K2**	07
	B. What are the features of Organisational Behaviour	**CO4**	**K2**	**07**
	OR			
8.	A. What is Motivation and Explain its importance	**CO4**	**K2**	07
	B. Explain about Maslow's Need Hierarchy theory	**CO4**	**K2**	**07**
	UNIT-V			
9.	A. What are the Ethics of Advertising	**CO5**	**K2**	07
	B. Explain about Corporate Social Responsibility	**CO5**	**K2**	**07**
	OR			
10.	A. What is Stress & Describe about causes of Stress	**CO5**	**K2**	07
	B. Identify the best practices to overcome Stress at organisational level	**CO5**	**K2**	**07**

					Course Code:B20HS2203
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R20**
II B. Tech II Semester - MODEL QUESTION PAPER					
BUSINESS COMMUNICATION AND VALUE SCIENCE-III					
Computer Science & Business System (CSBS)					
TIME: 3 Hrs.					**Max. Marks: 70 M**
Answer any One Question from Each Unit					
All questions carry equal marks					
			CO	**KL**	**M**
		UNIT-I			
1.	**a).**	Explain SWOT analysis and what are the basic principles of SWOT?	CO1	K3	7
	b).	Write an essay on how motivation helps develop your career.	CO1	K3	7
		OR			
2.	**a).**	What do you opine on Maslow's Theory about Motivation?	CO1	K3	7
	b).	Analyse the differences between SWOT and TOWS.	CO1	K4	7
		UNIT-II			
3.	**a).**	Draft an essay on Pluralism in cultural spaces?	CO2	K2	7
	b).	What are Glocal and Translocational impacts?	CO2	K2	7
		OR			
4.	**a).**	Discuss the theory of Pluralism in cultural spaces	CO2	K3	7
	b).	Define Cross-cultural communication and discuss how it is related to business communication.	CO2	K2	7
		UNIT-III			
5.	**a).**	Rewrite the following sentences. i. The major along with his soldiers was killed in the field. ii. They appointed him as a manager as he is efficient. iii. Modern film techniques are far superior than that employed in the past. iv. The three last chapters of this book are very interesting. v. No sooner the news had appeared in the paper than there was a rush in the counter. vi. I lived in a three hundred-years old house in Bombay. vii. A large number of houses are coming up in our town.	CO3	K2	7
	b).	What are the Dos and Don'ts of a Group Discussion.	CO3	K2	7
		OR			
6.	**a).**	Write the meanings of the following Idioms and Phrasal Verbs with one example sentence each. i. Hot cake ii. Take after iii. In hot water iv. Blue blood v. Call off	CO3	K2	7

		vi. Come across vii. Yellow-bellied			
	b).	Discuss types of grammar and their importance in constructing structurally correct sentences.	CO3	K3	7
		UNIT-IV			
7.	**a).**	Assume that you are an executive body member of an Entrepreneurship club of your college. Write a Report on the budget you spent for the Mock Summit of Entrepreneurship you organized in your college.	CO4	K3	7
	b).	What is Technical Writing and what are the basic principles of Technical Writing?	CO4	K2	7
		OR			
8.	**a).**	Make an Ad creating awareness of Omicron.	CO4	K4	7
	b).	Write an essay on Technical Writing and various types of Technical Writing.	CO4	K2	7
		UNIT-V			
9.	**a).**	Discuss the Role of Science in Nation Building.	CO5	K3	7
	b).	Define a Presentation and discuss the Dos and Don’ts of a Presentation.	CO5	K2	7
		OR			
10.	**a).**	Develop your views on the Role of Science Post-Independence.	CO5	K4	7
	b).	Write your plan for your presentation on “Voice of Future”.	CO5	K3	7